

SUKHAYA, Russian Federation

WMO: 307260

Lat: 52.552N

Lon: 107.106E

Elev: 1512

StdP: 13.91

Time Zone: 8.00 (E08)

Period: 86-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.9	-14.8	-24.9	1.5	-19.3	-19.6	2.0	-14.5	22.6	2.2	20.6	2.9	5.5	140	0.739

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	69.9	63.9	67.8	62.0	65.8	60.6	64.8	68.9	63.0	66.8	61.4	65.0	8.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	91.2	67.8	61.3	85.6	65.9	59.5	80.4	64.0	30.8	69.1	29.4	67.1	28.2	65.1	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	20.4	18.0	-23.1	79.0	6.7	4.1	-27.8	82.0	-31.7	84.4	-35.5	86.7	-40.3	89.7
			-23.4	66.9	5.8	2.2	-27.6	68.5	-31.0	69.8	-34.3	71.0	-38.6	72.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.5	2.4	1.7	13.8	29.5	41.5	51.2	58.4	57.9	47.7	36.5	22.0	13.2		
	DBStd	20.89	8.89	10.00	10.07	6.23	5.94	4.88	4.63	4.13	5.42	6.00	7.97	6.97		
	HDD50	7390	1476	1351	1123	615	272	44	2	2	106	420	840	1139		
	HDD65	12239	1941	1771	1588	1065	730	414	210	222	519	885	1290	1604		
	CDD50	634	0	0	0	0	7	80	262	247	37	1	0	0		
	CDD65	8	0	0	0	0	0	0	5	3	0	0	0	0		
	CDH74	10	0	0	0	0	0	0	7	3	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	7.9	7.4	6.7	7.1	7.6	7.7	7.7	7.2	7.1	8.4	8.9	9.8	9.5		
(15) Precipitation	PrecAvg	14.0	0.5	0.3	0.3	0.6	1.1	2.0	2.9	2.9	1.5	0.6	0.7	0.6		
	PrecMax	17.2	1.9	0.6	0.9	1.2	2.0	4.3	5.8	4.8	3.3	1.5	1.2	1.8		
	PrecMin	11.0	0.1	0.0	0.0	0.2	0.3	0.8	1.5	1.3	0.6	0.1	0.3	0.2		
	PrecStd	1.7	0.3	0.1	0.2	0.2	0.4	0.8	1.0	1.1	0.7	0.3	0.2	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	32.1	40.6	48.5	64.2	68.8	73.8	72.3	64.4	55.3	42.1	32.8		
		MCWB	26.5	27.6	34.0	40.9	51.8	58.6	66.3	64.9	58.4	47.7	35.7	28.7		
	2%	DB	25.1	23.8	35.8	44.0	58.6	64.7	70.7	69.4	60.9	51.1	38.3	29.4		
		MCWB	22.4	21.1	31.3	37.1	49.6	56.9	64.7	63.6	56.1	45.0	33.2	26.0		
	5%	DB	19.7	20.5	32.7	41.2	54.9	62.0	68.7	67.3	58.6	48.3	35.8	26.5		
		MCWB	17.4	18.3	28.9	35.6	47.2	55.6	63.2	62.0	54.4	43.1	31.3	23.3		
	10%	DB	16.2	17.0	29.6	38.8	51.7	59.8	66.7	65.3	56.3	45.8	33.1	23.6		
		MCWB	14.5	15.2	26.6	33.9	45.0	54.3	61.8	60.7	52.2	41.3	29.1	20.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	28.1	34.8	41.4	53.0	61.4	67.9	66.8	59.7	48.4	36.7	29.0		
		MCDB	29.3	32.0	39.6	48.2	59.8	66.2	72.1	70.6	62.5	53.9	40.3	32.2		
	2%	WB	22.4	21.3	31.8	37.6	50.5	58.4	65.6	64.5	56.7	46.0	33.7	26.0		
		MCDB	25.6	23.8	35.3	42.6	57.6	62.5	69.9	68.5	60.0	50.1	37.8	29.1		
	5%	WB	17.5	18.3	29.1	36.0	47.9	56.6	63.8	62.8	54.8	43.7	31.7	23.4		
		MCDB	19.6	20.3	32.4	40.7	54.1	60.7	67.7	66.5	58.0	47.6	35.4	26.0		
	10%	WB	14.5	15.2	26.5	34.2	45.4	54.9	62.2	61.2	52.9	41.7	29.3	20.9		
		MCDB	16.3	17.0	29.5	38.2	51.1	58.9	66.0	64.7	56.0	45.4	33.0	23.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	15.6	16.2	12.9	14.1	13.8	11.3	11.2	11.9	12.7	11.7	12.3		
		MCDBR	15.4	18.3	16.4	15.8	20.5	17.1	14.8	14.5	13.4	15.3	13.9	14.3		
	5% WB	MCWBR	13.6	16.2	12.9	11.2	12.9	11.1	9.8	9.8	9.9	11.3	11.4	12.6		
		MCWBR	15.4	17.8	15.6	14.3	18.6	14.8	12.9	13.1	12.2	14.1	13.6	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.292	0.322	0.367	0.427	0.444	0.435	0.387	0.337	0.310	0.283	0.266		
	taud		2.102	2.090	2.146	2.268	2.131	2.197	2.267	2.388	2.476	2.427	2.186	2.072		
	Ebn at Noon		224	254	272	274	261	258	258	266	269	252	216	197		
	Edh at Noon		23	32	37	38	46	43	40	34	27	23	21	20		
(44) All-Sky Solar Radiation	RadAvg		365	691	1142	1531	1822	1932	1788	1506	1115	686	362	248		
	RadStd		22	40	55	78	123	122	112	68	96	53	28	18		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (15 neighbors)		N/A	N/A	-3.20	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page